Revision of PAM50 Protein List for Breast Cancer Prediction Using Different Machine Learning Algorithms

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Abstract

Breast cancer is the most diagnosed cancer and the second common cause of cancer mortality worldwide. Classification of breast tumor stages is of vital importance for the selection of an adequate treatment. Recent advances in multi-omics technologies have made it possible to measure the expression of the complete proteome of a tumor sample. A collection of machine learning algorithms are utilized nowadays to classify breast tumor stages by using the proteome. In this survey several algorithms were applied on a proteome breast cancer dataset alongside the evaluation of the PAM50 gene classification set. We have found that the naive Bayes algorithm allows for the most accurate classification and that the PAM50 gene classification set was the best set to classify the dataset.

1 Introduction

According to the World Health Organization, breast cancer is the most common diagnosed cancer and is the second most common cause of cancer mortality worldwide (Ferlay *et al.*, 2012). Due to the severe heterogeneity in cancer tumors it is important to uncover the stage of the cancer which allows for the administration of an adequate treatment. Due the recent advances in multi-omics technologies, it is now possible to measure the expression of all proteins of a tumor sample. However, the expression levels of some specific genes are known to accurately indicate the stage of a cancer. Therefore, the ongoing search for these potential biomarkers is key in the treatment of breast cancer.

Breast cancer classifiers are used in medicine and research to accurately describe cancer subtypes. Each classifier uses a different spectrum and is based on different criteria. For example, the TNM classification system classifies tumors in size, using a range from T1-T4. T4 indicates a metastatic cancer which is also denoted by an additional TNM classifier (M0: no metastasis and M1: metastatic cancer). The most common classifier today classifies four molecular subtypes: luminal A, luminal B, HER2-enriched and basal-like (Perou *et al.*, 2000). This molecular subtype classifier is based on the expression of cell receptor. A set of 50 classifier genes (PAM50) were found which can potentially accurately determine the molecular subtype of a breast tumor.

Nowadays, machine learning (ML) algorithms are used to constitute a model that classifies cancer subtypes based on labeled high throughput data (supervised learning). There are two types of machine learning; supervised learning and unsupervised learning, which both are used in this study. In supervised learning, a label (i.e. cancer subtype) is assigned to each object in the dataset. This label can be used subsequently to extract classification rules from this earlier data. By using techniques like cross-validation, it is also possible to use these labels to validate the accuracy of the model. In unsupervised learning no labels are used. Therefore, unsupervised learning can be applied to explore the data and can suggest a specific number of classifications (Kourou *et al.*, 2015).

A well-known example of an unsupervised learning algorithm is Principal Component Analysis (PCA). PCA determines the spread of multivariate data by performing an orthogonal linear transformation, transforming the data in a new matrix, where the first component of this matrix shows the greatest variance of the data. This procedure is iterated to reduce the dimensions in the data leading to obtain an approximation of the variance in the data (Wold, 1987).

In the article of Boostani *et al.* (2017) several classification methods were compared to investigate which one has the highest prediction accuracy to select predictive biomarkers. It was shown that the Random Forest (RF) classifier outperformed the Gaussian Mixture Model, K-nearest-neighbour, Linear Discriminant Analysis with Nearest Center, and was therefore considered a good model to predict breast cancer biomarkers for classification. A RF is a classification method that selects several biomarkers. It consists of an ensemble of different decision trees, of which each tree is trained on a random subset of the data, followed by bootstrap selections. Selecting random training sets minimizes the variance of the final predictive model, resulting in more accurate results. Each tree gives an individual predictor as output, which altogether results in biomarker selections that are most abundant among the random trees (Breiman, 2001).

Besides RF, both Bayes and naive Bayes classifiers are used in classification problems. aive Bayes is a simple probability classifier based on the Bayes theorem, which uses the knowledge of conditions to describe the probability of a specific event. The model defines different problem instances by assigning class labels, in the form of vectors. All naive Bayes classifiers assume a strong independence between the different attributes whereas Bayes classifiers assumes a dependency between different attributes. Bayes classifiers are known to be efficient learners over small datasets, such as our dataset of 83 samples. Nevertheless, according to literature those classifiers are outperformed by RF and other bagging trees (Caruana, 2006).

Moreover, as a biomarker dataset typically contains an extremely large set of variables (genes) compared to the number of objects (samples) it has to endure the curse of dimensionality (Belman, 1961). Each variable adds a dimension to the system and causes an exponential increases in the number of possible combinations. In order to make a statistically validated representation of all those possible combinations, a large set of objects would be needed. RF addresses this problem by creating small subsets of randomly selected data and thereby reducing the number of variables tremendously.

To assess the representativeness of the constituted model, samples from the data set are randomly separated into two sets; training data and test data. The training set is used to train the model after which the accuracy of the model is evaluated by the test data. Cross-validation is the most common method that runs multiple validations in sequence. In each validation, a new training set and test set is chosen randomly after which the performance is averaged over all models. However, during cross-validation errors in both misclassification and generalization are produced. An increase in generalization results in the overfitting phenomenon which indicates that the model is not capable to classify new data correctly. Subsequently, decision trees are known to overfit datasets, i.e. have low bias and high variance. In contrast, RF is trained on different parts (i.e. minimizing the variance) and therefore has a slightly increased bias, but a greatly improved prediction model. Therefore, RF is a proper method to analyze proteome data sets in which the number of errors are kept at a minimum resulting in a convenient mode (Qi, 2013).

In this study, we evaluated supervised classification algorithms to determine the best possible prediction of PAM50 mRNA/molecular subtype class using the least number of proteins differently expressed with the highest accuracy, as found with unsupervised learning. As a baseline, we performed RF analysis with 10-fold cross-validation on the whole adjusted proteome to determine the highest possible accuracy.

For multiple reasons it is interesting to reduce the amount of proteins needed for prediction, for example it is much cheaper to do a lab experiment on 100 proteins than on 8000. We used principal component analysis to determine latent variables, known as principal components. Those principal components are then crossed with the data to find out which proteins contribute the most to that principal component in the dataset. Finally, 100 biomarkers (proteins) were selected based on the high contribution to the variance in both PC1 and PC2. The two new datasets are then tested on their accuracy for predicting the cancer subtypes. Besides those two additional datasets, an already known subselection (PAM50) is tested as well.

2 Methods

2.1 Dataset

In this survey, a breast cancer proteome dataset from the Clinical Proteomic Tumor Analysis Consortium (NCI/NIH) was used (Mertins et al, 2016). This dataset consists of the proteome data and the clinical data of 111 samples, including 25 basal-like, 29 luminal A, 33 luminal B, and 18 HER2 (ERBB2)-enriched tumors, along with 3 normal breast tissue samples. A total of 15.369 proteins (12.405 genes) were identified and measured on a numeric scale for 110 samples. 27 of these samples did not pass the quality requirements and were left out from the proteome dataset. Moreover, the dataset contains three biological replicates, these were regarded as novel entries to preserve the small number of samples in the dataset, resulting in 83 samples and 12.553 proteins.

Next to the clinical and proteome data, a list of genes from the Prediction Analysis of microarray 50 (PAM50) set was included (Parker *et al.*, 2009). This list contains 101 protein IDs that cover all reference proteins IDs that are derived from the 50 PAM50 genes. Besides these datasets, we wanted to make our own sub selections, which is done using principal component analysis.

2.2 Sample treatment

The proteomes were measured by iTRAQ, a novel high-throughput technique that measures the expression of proteins (Wiese *et al.*, 2006). This technique labels the peptides in the sample, after which liquid chromatography is performed to fraction all peptides. To measure the expression of each peptide, tandem mass spectrometry (MS/MS) was performed. MS/MS separates each peptide from the sample and measures the ionization of each particle, thereby obtaining the fragmentation data. By searching a database with reference fragmentation profiles, the labeled peptides can be identified and linked to the corresponding proteins. This resulted in the expression of each sample's proteome.

2.3 Data preprocessing

Outliers were identified with intensity plots of both proteins and samples, and were subsequently removed. Also, the proteins with missing values for at least one sample were also discarded. The experiment was continued with 83 samples and 8017 proteins. The normality of the data was evaluated by plotting the distribution of protein expressions in density plots for each of the samples. This indicated that auto-scaling was needed for normalization. All data preprocessing was performed using the *R programming language* (R Core Team, 2016) with the packages *affy* and *preprocessCore* from *Bioconductor* (Huber *et al.*, 2015).

2.4 Subselection

Two sub sections of the dataset were build by performing a PCA yielding a set of 10 principal components. These principal components were analyzed to determine which of the proteins in the dataset contributed the most to the variation in the queried principal component. The 100 most contributing proteins were selected to form the new dataset. This was done for both the first and the second component, resulting in two new datasets: PCA_1 and PCA_2.

2.5 Models

Four different classifiers are used on all datasets with *Weka*. These include: RF, Decision tree (J48), Bayes Classifier and naive Bayes. All ML algorithms and cross-validation were performed using *Weka* (Eibe, Hall and Witten, 2016). The scripts are attached in the supplementary material.Note that we ran a larger amount of machine learning methods including boosting, bagging, lazy learners and clustering, but those results could not exceed RF or naive Bayes and were therefore excluded from the result/discussion section.

3 Results & Discussion

3.1 Data preprocessing

Data was analyzed globally for outliers and whether it was normally distributed. In order to identify outlying sample samples, the intensities of metabolites were plotted for each of the 83 samples *(figure 1.A)*. For outlying metabolites, these intensities were plotted for the 12.553 metabolites *(figure 1.B)*. No samples or metabolites showed a consistent disordered pattern and therefore there were no outliers identified. Furthermore, it can be concluded that the data is already mean centered, as the way of measuring already gives intensities which are higher or lower than zero. Notably, the variability of samples tends to increase for metabolite ID numbers higher than 85.000, this group contains several different protein classes.

The majority of machine learning algorithms requires a dataset that does not contain any missing values. The amount of missing values in the dataset were identified in plots shown in *figure 2*. It seems that the amount of missing values increases with the metabolites ID number, and there is a large amount of missing values in the samples 75 to 80. Since the number of samples is quite small (n=83), we decided to preserve this information and exclude only metabolites from the large metabolite dataset (n=12.553), which after deletion was reduced to 63,9% of the original dataset (n=8017). In order to evaluate whether the protein expression of each sample was normally distributed, density plots were created for all samples, indicating that auto-scaling was needed for normalization (*figure 3 and supplementary material*).

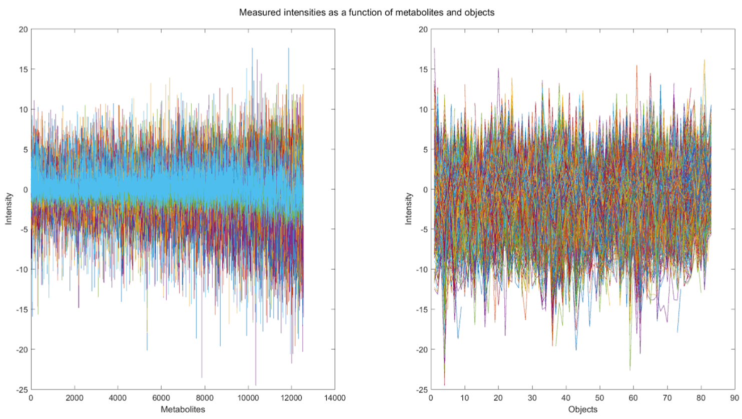

Figure 1: **Metabolite intensity plots show normal distribution and no outliers.** A) Each sample is plotted in a line representing its metabolite intensities. B) Each metabolite is plotted in a line for all samples.

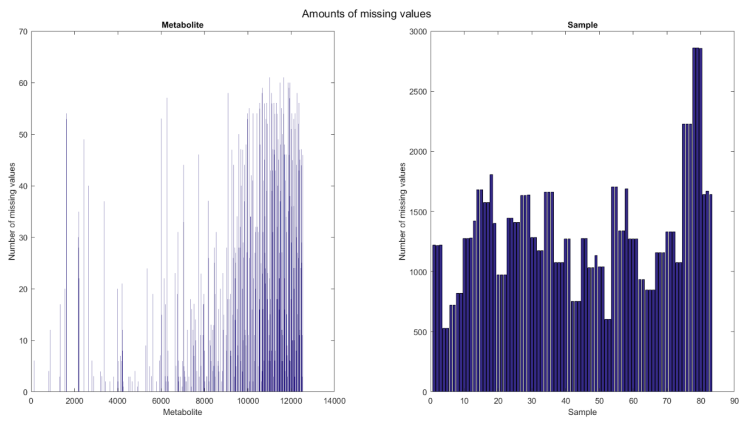

Figure 2: **Representation of the number of missing values for each metabolite (left) and each sample (right).**

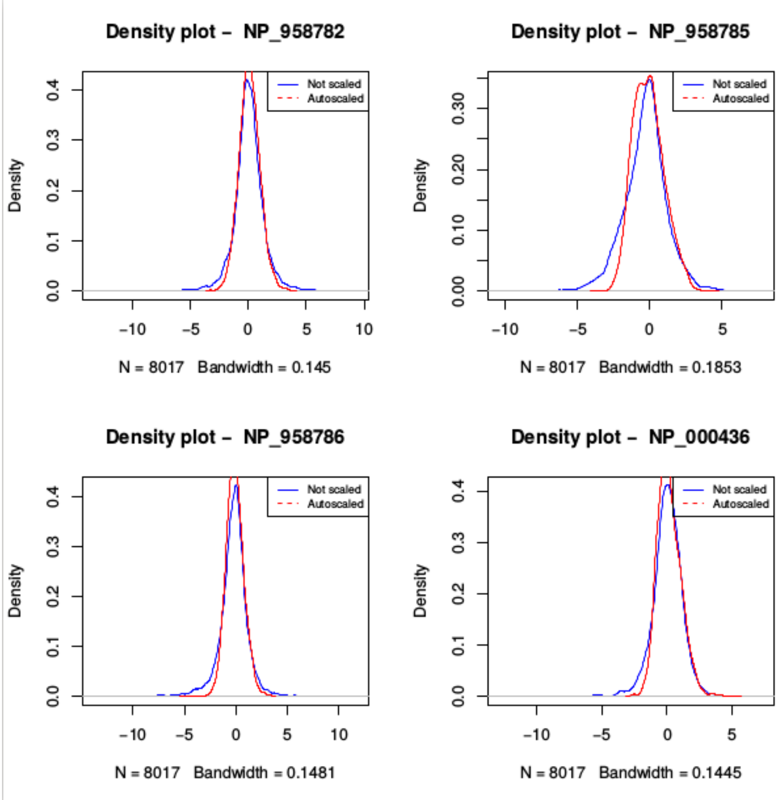

Figure 3: **Representation of four density plots of four randomly chosen samples.**

3.2 PCA on the most up and down regulated genes

Subsequently, the 50 most upregulated and 50 most downregulated genes were identified using PCA. From visualizations of components 1-7, it was concluded that component 2 was most discriminative to classify the proteome of a sample into one of the four PAM50 subclasses luminal A, luminal B, basal-like and HER2-enriched (*figure 4*). However, no component was found that clearly separated for all four subclasses.

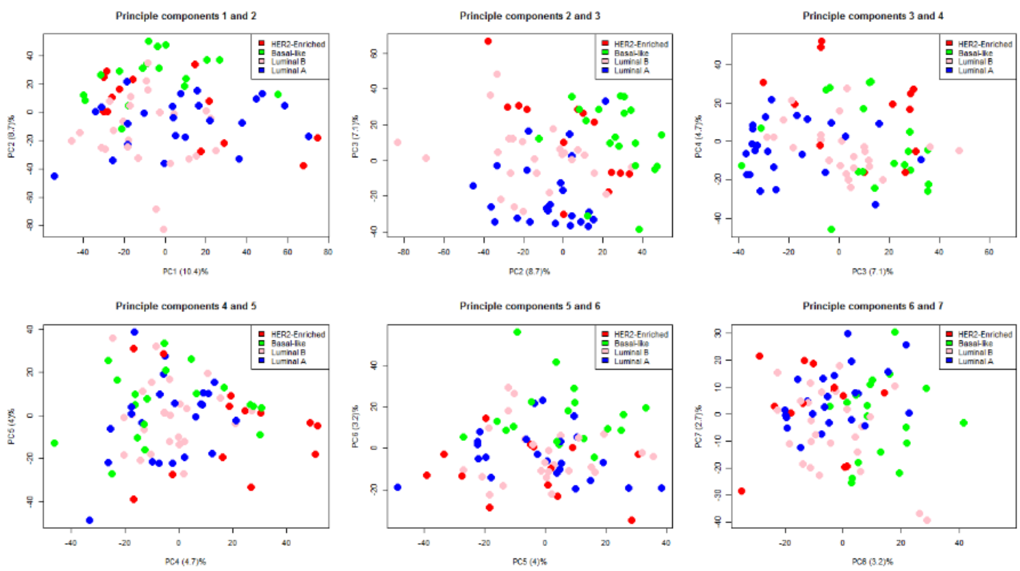

Figure 4: **Assessment of the principal component analysis.** Six plots representing the variance explained by the first 7 components.

A total of four different data sets were tested. First of all, the complete data set corrected for missing instances. Second, the dataset containing the most important proteins according to the Kaggle website. Third, the dataset derived from the first principal component and the dataset from the second component. Finally, four different classifiers were used on all those datasets using 10-fold cross-validation. A Random Forest, Decision tree (J48), Bayes and naive Bayes classifier were used. The final accuracies of all the models are shown in *table 1.*

Table 1: **Overview of the accuracy of each prediction method.** *Full Set* indicates the dataset containing 8017 proteins on 83 samples. *PCA*_100* are the 100 proteins selected based upon PC 1 and 2. *PAM50_genes* is the selection proposed on the Kaggle competition.

Classifier	Acc. Full Set	PCA1_100 Set	PAM50_genes	PCA2_100 Set
Random Forest	70.00	56.25	70.00	61.25
Decision tree	56.25	37.50	62.50	51.25
Bayes Classifier	67.50	48.75	62.50	48.75
Naive Bayes	68.75	62.50	76.25	62.50

The data selection from the PAM50 classifier *(PAM50_genes)* outperformed all the other datasets with a maximum prediction accuracy of 76%. The *PAM50_genes* dataset is based upon the PAM50 genes selected by Bernard *et al.* (2009). They selected these proteins IDs by performing a supervised Classification of Nearest Centroid algorithm. This is in contrast with our selection, which was based on unsupervised PCA using PC1 and PC2. Therefore these datasets are purely based on the variation between the samples and not necessarily on the cancer types. Seeing the lower prediction accuracy on the PCA datasets we expected the variation within the groups to be larger than between the groups, making it hard to distinguish between the multiple cancer types using PCA alone.

It is clear from table 1 that Random Forest and naive Bayes are the best classifiers for the full dataset, 70.00% and 68.75%, respectively. In advance we expected to see a better result with Random Forest, however, the naive Bayes classifier performed surprisingly well, out-classifying RF on 3 out of 4 cases. This could be explained as RF known to perform better with a higher number of samples, which, in our case is very limited (n=83). This is in contrast with the naive Bayes, which is a strong performer on datasets with relatively small sample sizes. Another reason could be that naive Bayes is more sensitive to overfitting than Random Forest and therefore looks like it only outperforms on the training set.

3.3 More samples and information discussion

Regarding that the classification was performed on only 83 samples, it is not surprising to retrieve a maximal prediction accuracy of 70%. We expect that the prediction accuracy can be improved by adding more samples. Which makes it easier for the Random Forest classifier to identify differences between the cancer classes. Moreover, to reduce the amount of attributes while maintaining the prediction accuracy, selecting more biological properties would yield better results. The allowed variance selection between the samples was not capable to increase the prediction accuracy. However, both the PCA selection sets performed poorly in contrast to the biological variant (*PAM_50*). It is clear that a future selection should be more focused on the differences between the groups rather than the samples.

3.4 Settings used in R & Weka

Throughout this survey, different settings were used for all machine learning algorithms to improve the classification accuracy. Unfortunately, none of these alterations improved the accuracy.

For all the classifiers default *Weka* setting were used. For Random Forest default iterations is set to 100, increasing the iteration did slightly improve the prediction accuracy, nevertheless this effect was negligible compared to the risk of overfitting, therefore we choose to continue with the default settings for a more general model. For naive Bayes the normal distribution was used rather than a kernel estimator, since the prediction accuracy decreased with the kernel estimator.

3.5 Further Research

Enriched Random Forest is a new method that has shown great potential in the analysis of proteome data (Dhammika *et al.* 2008). This method is particularly useful for data that is highly adaptive containing a big set of variables and a low number of objects. Subsequent research could make use of this method which could lead to an improvement of the prediction accuracy. Furthermore, a big collection of objects and genes with high number of missing values were excluded from the dataset in this survey. A PCA can be performed after which the yielded scores and loadings can be used to generate equivalent entries for these missing values. This method allows for a good approximation of the missing values and could have resulted in different results.

3.6 Overall conclusion

We conclude that for this small dataset of only 83 samples the naive Bayes classifier is a better predictor than RF. Although, we predict that RF would classify better on a larger dataset than naive Bayes, thereby improving the overall prediction accuracy. Also, this dataset underlines that the PAM50 classifier is a good classifier for breast cancer tumor samples.

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